

3.1.5 Mechanical devices

Different types of movement

Content	Potential links to maths and science
The functions of mechanical devices to produce linear, rotary, reciprocating and oscillating movements.	Visualise and represent 2D and 3D objects including 2D diagrams of mechanisms/mechanical movement.

Changing magnitude and direction of force

Content	Potential links to maths and science
Levers: • first order • second order • third order. Linkages: • bell cranks • push/pull. Rotary systems: • CAMs and followers • simple gear trains • pulleys and belts.	The action of forces and how levers and gears transmit and transform the effects of forces. Arithmetic and numerical computation eg use ratios. Use angular measures in degrees, visualise and represent 2D and 3D objects including 2D diagrams of mechanisms/mechanical movement. Knowledge of the function of mechanical devices to produce different sorts of movement, changing the magnitude and direction of forces.